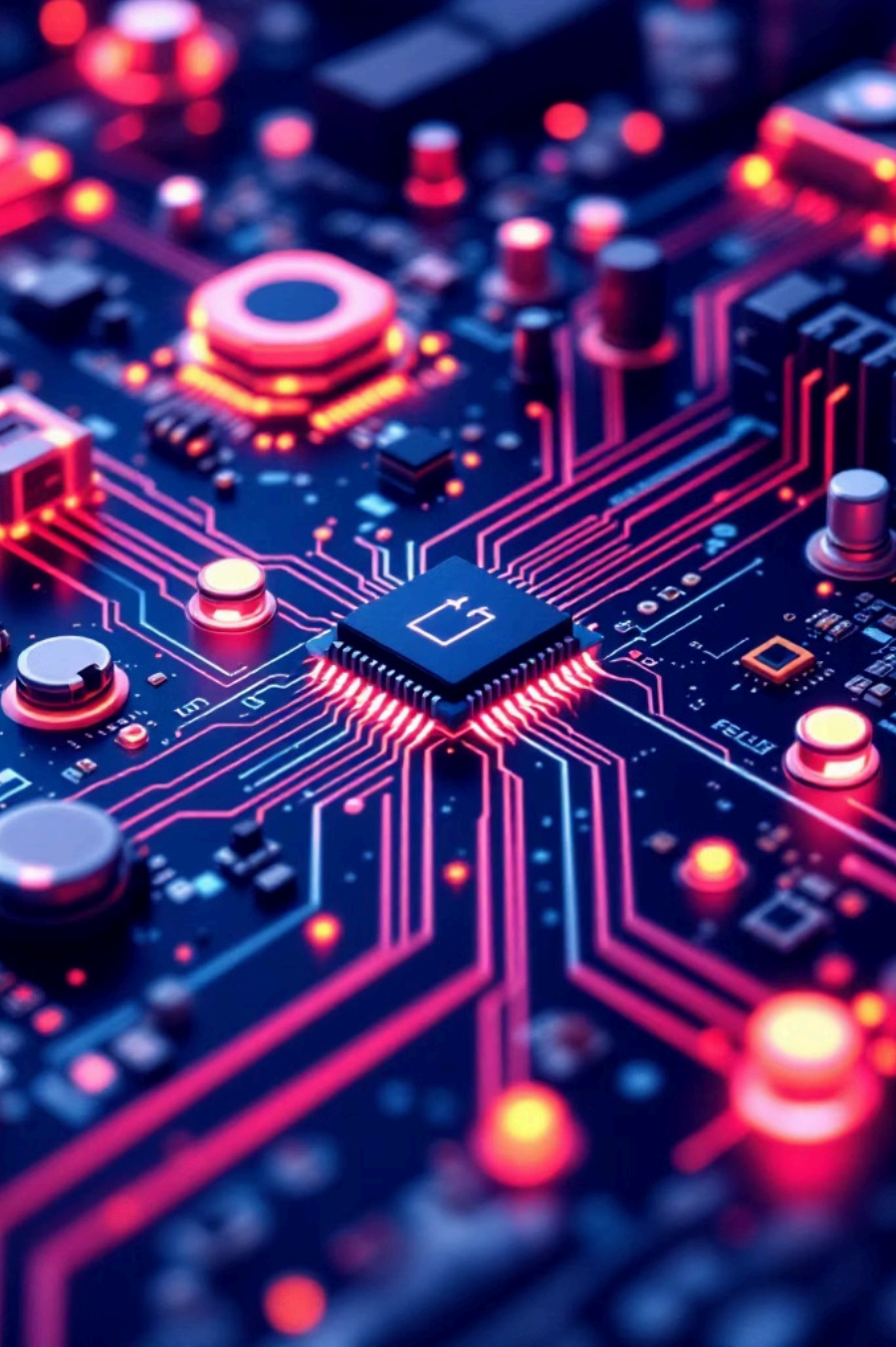


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🎓 Sensors and Actuators in IoT

Types, Technology Principles & Real-World Applications

Group Assignment | IoT & Embedded systems

Sensors in IoT (Overview)

What is a Sensor?

- A device that detects physical/environmental changes
- Converts physical quantities into electrical signals
- Provides input to IoT systems

Most Common IoT Sensors:



Temperature



Humidity



Motion



Light



Gas

 **Key Idea:** Sensors allow IoT systems to *understand the environment*.

Popular Environmental Sensors

1 Temperature Sensor

Example: DHT22, LM35

Technology:

- Resistance or voltage changes with temperature
- Digital sensors use built-in ADC

Application:

- Smart thermostats
- Greenhouse monitoring

2 Humidity Sensor

Example: DHT22

Technology:

- Capacitive sensing
- Moisture changes capacitance

Application:

- Smart irrigation
- Weather stations

Popular Detection Sensors

1

Motion Sensor

Example: PIR sensor

Technology:

- Detects infrared radiation changes from moving objects

Application:

- Security systems
- Automatic lighting

2

Light Sensor

Example: LDR

Technology:

- Resistance decreases as light increases

Application:

- Automatic street lights
- Phone brightness control

3

Gas Sensor

Example: MQ-2

Technology:

- Gas exposure changes semiconductor resistance

Application:

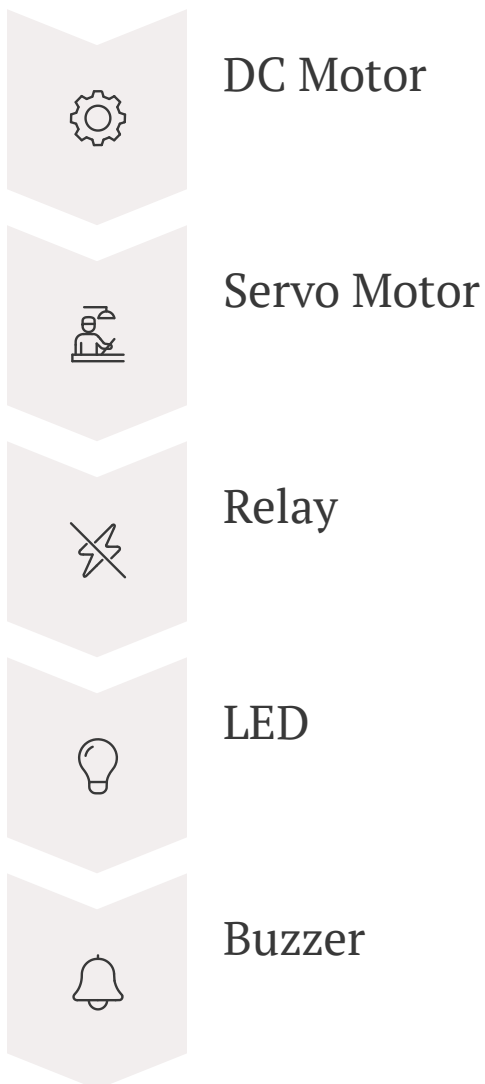
- Smoke detection
- Air quality monitoring

Actuators in IoT (Overview)

What is an Actuator?

- Converts electrical signals into physical action
- Produces movement, light, sound, or switching

Most Common IoT Actuators:



 **Key Idea:** Actuators allow IoT systems to *interact with the environment*.

Popular Actuators

1 DC Motor

Technology:

Electromagnetism →
electrical energy into
rotation

Application: Smart
doors, conveyor
systems

2 Servo Motor

Technology: PWM +
feedback system for
precise position

Application: Robotic
arms, camera control

3 Relay

Technology:

Electromagnetic switch
controlling high voltage

Application: Smart
home appliance control

4 LED / Buzzer

Technology:

- LED →
Electroluminescence
- Buzzer →
Piezoelectric effect

Application:

- Status indicators
- Alarm systems

How They Work Together

IoT Workflow



This workflow illustrates the complete cycle from sensing to action in any IoT system.

Example:

Temperature rises → Controller detects → Fan motor turns ON



Short Conclusion

Sensors = Input

Actuators = Output

Together they enable automation, monitoring, and smart control

Applications include:



Smart homes



Smart agriculture



Industrial automation